

Clinical Decision Support System

MRI Brain Tumor Analysis Report

Patient ID	ANON_PITUIT
Date/Time	2026-05-18T09:58:15.497194
Run ID	20260518_095810
Tumor Type	PITUITARY
Confidence	100.0%
Radiologist	Approved

Clinical Summary

Pituitary adenoma detected. Endocrine workup, visual field testing, and neurosurgery or endocrinology consultation are recommended depending on tumor size and functional status.

Recommended Next Steps

1. Endocrinology consultation
2. Full hormone panel (prolactin, IGF-1, cortisol, TSH, LH/FSH)
3. Formal visual field testing (Humphrey perimetry)
4. Dedicated pituitary MRI protocol if not already done
5. Neurosurgery referral if macro-adenoma or causing visual symptoms

Clinical Guidelines

PITUITARY ADENOMA

Definition: Pituitary adenoma is a benign neoplasm of the anterior pituitary gland arising from secretory cells. It is the most common intrasellar mass.

Characteristics:

- Accounts for approximately 15% of primary intracranial tumors
- Microadenoma: < 10 mm diameter; Macroadenoma: ≥ 10 mm
- Functional adenomas secrete hormones (prolactin, GH, ACTH, TSH, FSH/LH)
- Non-functional adenomas cause symptoms via mass effect
- Pituitary apoplexy (hemorrhage/infarction) is a neurosurgical emergency

Typical Imaging Findings (MRI):

- Rounded lesion within the sella turcica
- Homogeneous signal on T1 (isointense to gray matter)
- Homogeneous enhancement with gadolinium (delayed compared to normal pituitary)
- Macroadenomas may extend suprasellarly, compress optic chiasm, or invade cavernous sinus
- Pituitary stalk deviation for microadenomas
- Cystic/necrotic change possible in larger tumors

Recommended Next Steps:

1. Endocrinology consultation and comprehensive hormone panel:
 - Prolactin, IGF-1 (for acromegaly), 24-hour urine free cortisol (for Cushing's)
 - TSH, free T4, LH, FSH, testosterone/estrogen
2. Formal visual field testing (Humphrey perimetry) if suprasellar extension suspected
3. MRI with dedicated pituitary protocol (thin 2-3 mm slices, dynamic contrast)
4. Neurosurgery referral for macroadenomas causing mass effect or visual symptoms
5. Serial MRI surveillance (every 6-12 months) for stable microadenomas
6. Cabergoline/bromocriptine for prolactinomas (first-line medical therapy)

Prognosis:

- Generally excellent — most are benign with good outcome after treatment

- Surgical cure rates: ~70-80% for microadenomas, ~50% for macroadenomas
- Medical cure possible for prolactinomas and GH-secreting tumors
- Hypopituitarism may persist post-operatively requiring lifelong replacement

Treatment:

- Prolactinoma: dopamine agonists (cabergoline, bromocriptine) — first-line
- Other functional adenomas: transsphenoidal surgery (endoscopic or microscopically assisted)

This report is generated by an AI-assisted clinical decision-support system. It is NOT a substitute for professional medical diagnosis. All findings must be confirmed by a qualified radiologist or physician.